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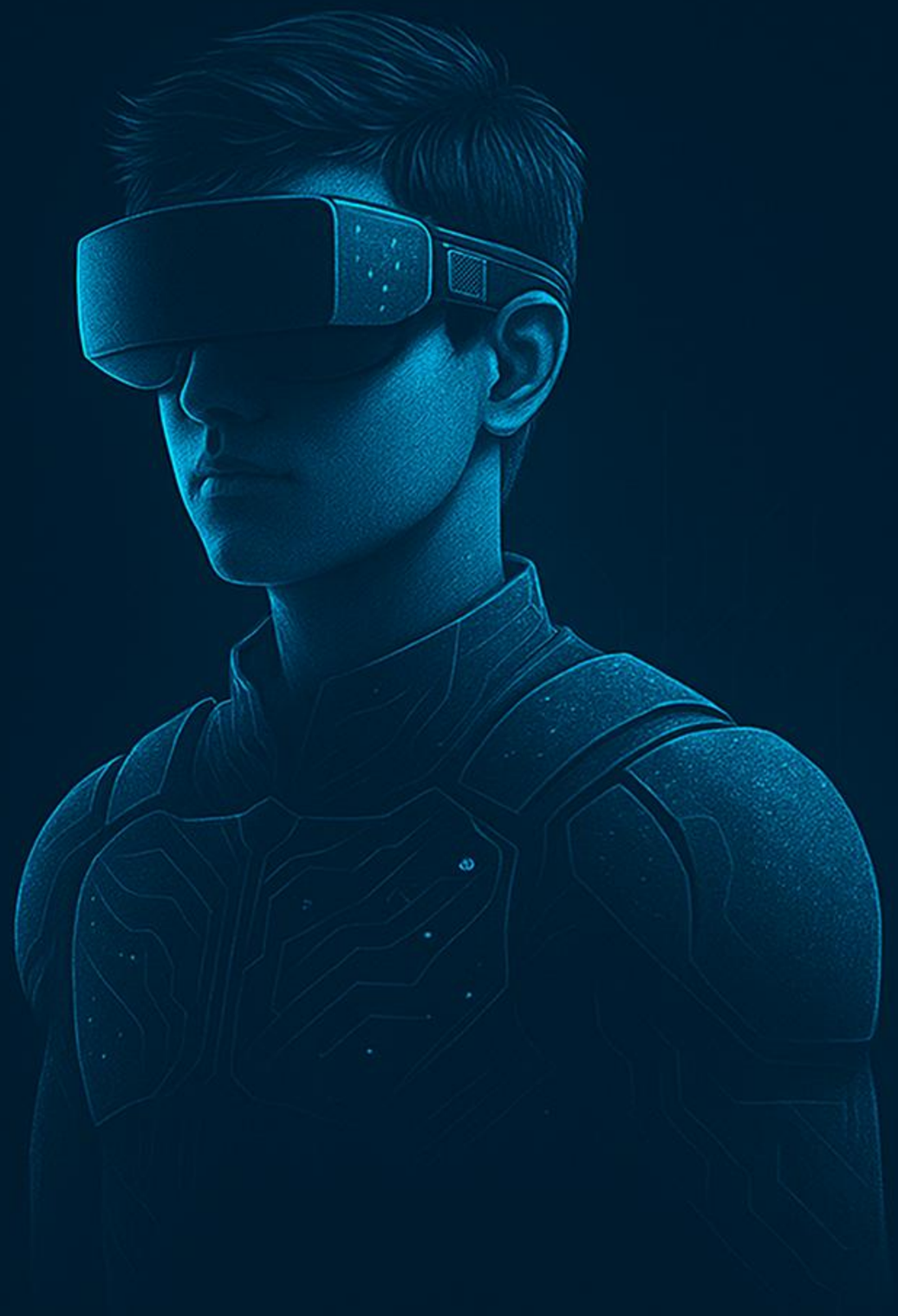
Module 6: Cybersecurity Fundamentals

Day 5 - SIEM Fundamentals



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Module 6 — Course Overview

Day	Date	Topic
✓ Day 1	Mar 18	Threat Intelligence & Frameworks
✓ Day 2	Mar 25	URL Analysis & Vulnerability Fundamentals
✓ Day 3	Apr 15	Phishing & Email Security
✓ Day 4	Apr 22	Password Security & Network Traffic Analysis
▶ Day 5	**Apr 29**	**Malware Analysis & SIEM**
Day 6	May 6	Incident Response Reporting
Day 7	May 13	Capstone — Simulated Incident Investigation

What We Learned So Far — Days 1–4

- Day 1 — Threat Intelligence: Kill Chain, ATT&CK, OSINT, STIX/TAXII, ISACs
- Day 2 — URL & Vulnerabilities: Nmap, Nessus, CVEs, OWASP Top 10, Metasploit
- Day 3 — Phishing & Email Security: BEC, quishing, vishing, SPF/DKIM/DMARC
- Day 4 — Password Security & Network Traffic:
  - Password hashing, Hashcat, John the Ripper, PtH, Kerberoasting
  - Network traffic analysis, Wireshark, C2 detection, DNS tunneling, ARP spoofing

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What is Malware?

- **Malware:** Malicious software designed to harm, exploit, or compromise systems
- **Common types:**
  - **Virus:** Self-replicating, attaches to files
  - **Worm:** Self-propagating across networks
  - **Trojan:** Disguised as legitimate software
  - **Ransomware:** Encrypts data for payment
  - **Spyware:** Steals information
  - **Rootkit:** Hides presence, maintains access

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Malware Types — Know the Difference

Type	How It Spreads	Key Behavior	Famous Example
Virus	Attaches to files; user shares	Needs a host file to survive	ILOVEYOU (2000)
Worm	Network exploits; fully automatic	No user action needed	WannaCry (2017)
Trojan	Disguised as legit software	User installs it; opens backdoor	Emotet
Ransomware	Phishing, exploits, worms	Encrypts files; demands payment	LockBit, REvil
Spyware	Bundled / zero-click exploits	Records keystrokes, screens, creds	Pegasus (NSO)
Rootkit	Post-compromise installation	Hides in OS kernel; invisible	Stuxnet (2010)
RAT	Phishing, trojans, exploits	Full remote control of machine	Cobalt Strike

Malware can be multiple types simultaneously: WannaCry = Worm + Ransomware | Stuxnet = Worm + Rootkit | Emotet = Trojan + Loader

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Malware Analysis Goals

- **Why analyze malware?**
- **Identification:** What is this malware?
- **Capabilities:** What can it do?
- **Indicators:** How do we detect it?
- **Attribution:** Who created it?
- **Remediation:** How do we remove it?
- **Analysis informs defense**

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Analysis Types

Type	Method	Skill Level	Time
Automated	Sandbox submission	Low	Minutes
Static	Examine without running	Medium	Hours
Dynamic	Execute and observe	Medium-High	Hours
Code Analysis	Reverse engineering	High	Days

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Safe Analysis Environment

- **Safety requirements:**
  - –Isolated virtual machines
  - –No network connectivity (or monitored fake network)
  - –Snapshots before execution
  - –Separate from production network
  - –Host-based protections
- **Never analyze malware on production systems!**

Automated Sandbox Analysis

- **Online sandboxes:**
  - **Any.run:** Interactive analysis
  - **Hybrid Analysis:** Detailed reports
  - **VirusTotal:** Multi-engine scanning
  - **Joe Sandbox:** Comprehensive analysis
  - **Cuckoo Sandbox:** Self-hosted option (outdated!)
- **Submit sample → Receive report**

Sandbox Report Analysis

- **Report sections:**
  - **File information:** Hashes, type, size
  - **Detection:** AV engine results
  - **Network activity:** DNS, HTTP, connections
  - **File activity:** Created, modified, deleted files
  - **Registry activity:** Keys modified
  - **Process activity:** Spawned processes
  - **Extracted IOCs:** IPs, domains, hashes

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Static Analysis Basics

- **Static analysis examines files without execution:**
  - **Techniques:**
    - File type identification
    - Hash calculation
    - String extraction
    - Metadata analysis
    - Packer detection
    - Import/export analysis
  - **Tools:** file, strings, exiftool, PEStudio, DIE

String Analysis

- **Strings reveal:**
  - –URLs and IP addresses
  - –File paths
  - –Registry keys
  - –Error messages
  - –Commands
  - –Encryption keys (sometimes)
- **Linux/Windows:**

```
strings malware.exe | grep -i "http"
strings -a -n 6 malware.exe
```

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PE File Analysis

- **PE (Portable Executable):** Windows executable format
- **Analysis points:**
  - –Headers (compilation time, entry point)
  - –Sections (.text, .data, .rsrc)
  - –Imports (DLLs and functions used)
  - –Exports (functions provided)
  - –Resources (embedded files)
- **Tools:** PEStudio, PE-bear, pestudio

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Identifying Suspicious Imports

Function	Purpose
VirtualAlloc	Memory allocation (shellcode)
CreateRemoteThread	Code injection
WriteProcessMemory	Process manipulation
RegSetValueEx	Registry modification
InternetOpen	Network communication
CryptEncrypt	Encryption (ransomware)

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Packer Detection

- **Packers:** Compress/encrypt malware to avoid detection
- **Indicators of packing:**
  - –High entropy (randomness)
  - –Few readable strings
  - –Small import table
  - –Unusual section names
  - –Known packer signatures
- **Tools:** Detect It Easy (DIE), PEiD, Exeinfo PE

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Dynamic Analysis Basics

- **Dynamic analysis executes malware to observe behavior:**
- **What to monitor:**
  - –Process creation
  - –File system changes
  - –Registry modifications
  - –Network connections
  - –API calls
- **Tools:** Process Monitor, Procmon, Wireshark, API Monitor

Process Monitor (ProcMon)

- **Process Monitor:** Windows real-time monitoring tool
- **Captures:**
  - –File system operations
  - –Registry operations
  - –Network activity
  - –Process/thread activity
  - –**Filters:** Focus on malware process, exclude noise
- **Essential for Windows malware analysis**

Document Malware Analysis

- **Malicious documents:**
  - Office files with macros (DOC, DOCX, XLS)
  - PDFs with JavaScript or embedded objects
  - RTF with exploits
- **Analysis approach:**
  1. Extract macros/scripts
  2. Deobfuscate code
  3. Identify payload delivery
  4. Execute in sandbox if needed
- **Tools:** oletools, olevba, pdf-parser

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Analyzing Office Macros

- **olevba:** Extract and analyze VBA macros
- **Look for:** Auto-execution, shell commands, downloads

```
olevba malicious.doc

# Output shows:
# - AutoOpen/AutoExec (automatic execution)
# - Suspicious keywords (Shell, PowerShell, http)
# - Obfuscation indicators
# - Extracted macro code
```

Malware Analysis Workflow

- **Recommended workflow:**
  1. **Gather info:** Hashes, context, source
  2. **Automated analysis:** Submit to sandbox
  3. **Static analysis:** Strings, PE analysis, scripts
  4. **Dynamic analysis:** Execute and monitor
  5. **Document findings:** IOCs, capabilities, recommendations
  6. **Share intelligence:** Report to team/community

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Why Memory Forensics?

- **Memory contains:**
  - –Running processes (including hidden)
  - –Network connections
  - –Encryption keys
  - –Decrypted malware
  - –User credentials
  - –Command history
- **"Memory doesn't lie"**

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Memory Acquisition

- **Acquisition tools:**
  - **FTK Imager:** Free, reliable
  - **WinPMEM:** Open source
  - **Magnet RAM Capture:** Free
  - **LiME:** Linux Memory Extractor
- **Considerations:**
  - Acquisition changes memory
  - Time-sensitive (volatile)
  - Large files (GB)
  - Live system required

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Volatility Framework

- **Volatility:** Premier memory forensics framework
- **Capabilities:**
  - –Process listing (including hidden)
  - –Network connections
  - –Loaded DLLs
  - –Registry extraction
  - –Malware detection
- **Versions:** Volatility 2 (Python 2), Volatility 3 (Python 3)

Volatility Profile Selection

- **Profile = Memory structure definition**
- **Volatility 2:**
- **Volatility 3:**

```
volatility -f memory.dmp imageinfo
volatility -f memory.dmp --profile=Win10x64_19041 pslist
vol.py -f memory.dmp windows.info
vol.py -f memory.dmp windows.pslist
```

- **Volatility 3 auto-detects profiles**

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Essential Volatility Plugins

Plugin	Purpose
pslist/pstree	List processes
netscan	Network connections
malfind	Find injected code
dlllist	Loaded DLLs
handles	Open handles
cmdline	Command line arguments
hashdump	Extract password hashes

Finding Hidden Processes

- **Detecting hidden processes:**

```
# Normal process list (can be manipulated)
vol.py -f mem.dmp windows.pslist

# Scan for process structures (harder to hide)
vol.py -f mem.dmp windows.psscan
```

- **Compare results:** Processes in psscan but not pslist are hidden

Detecting Code Injection

- **malfind plugin:** Detects suspicious memory regions

```
vol.py -f mem.dmp windows.malfind
```

- **Indicators:**
  - Executable memory not backed by file
  - PAGE_EXECUTE_READWRITE permissions
  - Suspicious content (shellcode patterns)
- **Dumps suspicious regions for analysis**

Network Connection Analysis

- **Network artifacts in memory:**
- **Shows:**
  - Local and remote addresses
  - Ports and protocols
  - Connection state
  - Associated process

```
vol.py -f mem.dmp windows.netscan
```

- **Find C2 connections from memory**

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Memory Forensics Workflow

- **Analysis workflow:**
  1. **Acquire memory** (before shutdown!)
  2. **Identify profile/OS**
  3. **List processes** (pslist/psscan comparison)
  4. **Check network connections** (netscan)
  5. **Detect injection** (malfind)
  6. **Extract artifacts** (dump suspicious processes)
  7. **Document and correlate** findings

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What is a SIEM?

- **SIEM = Security Information and Event Management**
- **Core functions:**
  - –Log collection and aggregation
  - –Normalization and parsing
  - –Correlation and analysis
  - –Alerting and notification
  - –Investigation and search
  - –Reporting and compliance

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SIEM Data Flow

- **Log flow:**

```
Sources → Collection → Parsing → Storage → Analysis → Action

Endpoints  ]
Servers    ] → Forwarders → SIEM → Index → Rules → Alerts
Network    ]                  ↓
Applications]              Search/Dashboards
```

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Log Sources

Source	Value
Windows Events	Authentication, process, PowerShell
Linux Syslog	Auth, system, application logs
Firewall	Network allows/denies
Proxy/Web	URL access, downloads
DNS	Query logs for detection
EDR	Endpoint detection data
Cloud	AWS/Azure/GCP audit logs

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Windows Event Forwarding

- **Windows Event Forwarding (WEF):**
  - –Native Windows log collection
  - –Centralized subscription model
  - –No agent required
  - –Scalable to large environments
- **Key events to collect:**
  - –Security (4624, 4625, 4688, 4672)
  - –PowerShell (4103, 4104)
  - –Sysmon (if installed)

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Sysmon — System Monitor

- Free Microsoft Sysinternals tool — deep Windows activity logging
- Key Event IDs:
  - Event 1 — Process Create (full command line + parent process + hashes)
  - Event 3 — Network Connection (process name, src/dst IP, port)
  - Event 7 — Image Load (DLLs loaded — detect DLL hijacking)
  - Event 11 — File Create
  - Event 22 — DNS Query (detect DNS-based C2)
- Install: sysmon64.exe -accepteula -i sysmonconfig.xml
- Recommended config: SwiftOnSecurity or Olaf Hartong (GitHub)
- Output → Event Log: Applications and Services → Microsoft → Windows → Sysmon/Operational

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Critical Windows Event IDs

- Authentication:
  - 4624 — Successful logon | 4625 — Failed logon (brute force)
  - 4648 — Logon with explicit credentials | 4672 — Special privileges assigned
  - 4776 — NTLM authentication attempt
- Account Management:
  - 4720 — Account created | 4726 — Account deleted
  - 4732/4728 — Added to security group / Domain Admins
- Process / Execution:
  - 4688 — Process created (enable command-line logging in audit policy!)
  - 4104 — PowerShell script block logging
- Services / Persistence:
  - 7045 — New service installed | 4698 — Scheduled task created

SOAR — Security Orchestration, Automation and Response

- Extends SIEM: automates repetitive analyst tasks using playbooks
- Example playbook — Phishing Alert:
  - 1. Parse email → extract URLs, attachments, sender
  - 2. Submit to VirusTotal / sandbox automatically
  - 3. If malicious: block IP/domain at firewall, quarantine email
  - 4. Notify affected user | 5. Create incident ticket
- Tools: Splunk SOAR (Phantom), Palo Alto XSOAR, Microsoft Sentinel Logic Apps
- SOAR shifts analyst role: alert handler → playbook builder + exception reviewer
- Risk: automate only well-understood, high-confidence detections

Sigma Rules — Platform-Agnostic Detection

- Sigma: open standard for writing SIEM detection rules
- Write once → convert to any SIEM: Splunk, Elastic, QRadar, Sentinel
- Like Snort (network) or YARA (files) — but for log-based detection
- Rule structure: title | logsource (product/category) | detection | level
- Example: detect Mimikatz by process name + command line
- Repository: github.com/SigmaHQ/sigma — 5,000+ community rules
- Converter: pySigma (python) → generates native SIEM query
- Benefit: vendor-neutral — share detections across the entire community

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Syslog Collection

- **Syslog:** Standard Unix/Linux logging protocol
- **Components:**
- **Facility:** Log source category
- **Severity:** Message importance
- **Message:** Log content
- **Collection:**

```
# rsyslog forwarding
*.* @@siem.company.com:514
```

- **UDP (514) or TCP (514/6514)**

Log Parsing and Normalization

- **Parsing transforms raw logs:**
- **Raw:**

```
Jan 15 10:23:45 server sshd[1234]: Failed password for
admin from 192.168.1.100
```

- **Parsed:**

```
{
"timestamp": "2024-01-15T10:23:45",
"host": "server",
"service": "sshd",
"event": "failed_login",
"user": "admin",
"source_ip": "192.168.1.100"
}
```

Detection Rules and Correlation

- **Detection approaches:**
- **Single event:** One event triggers alert
- **Threshold:** Count exceeds limit
- **Correlation:** Multiple events combine
- **Sequence:** Events in specific order
- **Example correlation:**

```
IF failed_login count > 5 within 5 minutes
AND source_ip same
THEN alert "Brute Force Attempt"
```

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Alert Management

- **Alert lifecycle:**
  1. **Triggered:** Rule matches, alert created
  2. **Triaged:** Analyst reviews priority
  3. **Investigated:** Deep dive into evidence
  4. **Escalated:** If confirmed, incident created
  5. **Closed:** False positive or resolved
- **Key metrics:** Alert volume, false positive rate, time to triage

SIEM Platforms

- **Common SIEM platforms:**
  - **Splunk:** Market leader, powerful, expensive
  - **Elastic Security:** Open source core, scalable
  - **Microsoft Sentinel:** Cloud-native, Azure integration
  - **IBM QRadar:** Enterprise-focused
  - **LogRhythm:** Mid-market option
  - **Wazuh:** Open source, endpoint-focused

Splunk Basics

- **Splunk components:**
  - **Search Processing Language (SPL)**
  - **Indexes:** Data storage
  - **Sourcetypes:** Log format definitions
  - **Apps:** Extended functionality
  - **Dashboards:** Visualization
  - **Basic search:**

```
index=security sourcetype=WinEventLog EventCode=4625
```

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Elastic Stack (ELK)

- **Elastic Stack components:**
  - **Elasticsearch:** Search and storage engine
  - **Logstash:** Log processing pipeline
  - **Kibana:** Visualization and search UI
  - **Beats:** Lightweight data shippers
  - **Query language:** KQL (Kibana Query Language)
- **Open source + commercial features**

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Day 5 Summary

- Malware analysis: automated sandbox → static (strings, PE imports) → dynamic (ProcMon)
- Memory forensics: Volatility plugins — pslist, netscan, malfind, dlllist
- SIEM: centralised log collection, normalisation, correlation rules, alerting
- Sysmon: deep Windows logging — process create, network connect, DNS query (Events 1/3/22)
- Critical Windows Event IDs: 4624/4625 (logon), 4688 (process), 4104 (PowerShell), 7045 (service)
- SOAR: automates alert playbooks — block, notify, ticket, enrich
- Sigma rules: write once, convert to any SIEM platform

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Thank You